AI7101 Project: Pump It Up

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1 Introduction and data

1.1 Problem and intended use

Identifying waterpoints requiring attention can help prioritize scarce inspection and maintenance resources. The Pump It Up task classifies the recorded operating condition of Tanzanian waterpoints as functional, functional needs repair, or non-functional. Functional sites are operational without repair needs; repair sites remain operational but require repairs; non-functional sites are not operational [1].

This study compares eight classifier families and selects a model and repair-probability threshold using macro-F1. It then evaluates condition classification and an illustrative inspection policy. The policy treats actual repair and non-functional sites as attention cases, allowing inspection coverage to be distinguished from correct three-class labeling.

1.2 Dataset and class imbalance

The dataset contains 59,400 labeled records and 14,850 unlabeled test records, with 39 predictors describing location, elevation, funding, installation, management, construction, extraction technology, payment, water quality, quantity, source, and waterpoint type [1]. Inputs and targets are aligned by a unique identifier, which is excluded from prediction. Training and test identifiers are disjoint, and their predictor fields match.

Functional waterpoints make up 54.31% of the labeled data, whereas only 7.27% belong to the repair class (Table 1). A majority classifier can therefore attain approximately 54.31% accuracy without detecting any repair cases. Macro-F1 provides a more balanced basis for model selection.

Table 1: Class distribution in the labeled dataset and reserved holdout.

Class	Labeled records	Share	Holdout records
Functional	32,259	54.31%	3,226
Functional needs repair	4,317	7.27%	432
Non-functional	22,824	38.42%	2,282

1.3 Exploratory findings

Exploration is restricted to the 53,460 development records. Of 5,603 dry waterpoints, 5,424 (96.81%) are non-functional. Among 5,756 sites of type `other`, 4,745 (82.44%) are non-functional. These associations support retaining water quantity and waterpoint type and considering nonlinear classifiers, but do not establish causation.

Scheme name has 48.50% missing values, and several administrative variables have approximately 5–7% missingness. Sparse categories require caution: only six development sites have type `dam`. Geographic clustering also motivates validation across regions, since random partitions can place nearby sites in both training and validation sets.

2 Feature engineering and preprocessing

Feature engineering converts the 39 original predictors into 37 fields before categorical expansion. The transformations address inconsistent entries, redundancy, sparse identifiers, seasonality, and skew while preserving physical and administrative context (Table 2).

Table 2: Feature transformations and their rationale.

Transformation	Rationale and implementation
Remove redundancy	Drop the constant recording-organization field, the exact duplicate of water quantity, and selected overlapping grouped variants.
Reduce sparse identifiers	Remove raw waterpoint name, subvillage, and scheme name; preserve scheme-name availability with a missingness indicator.
Qualify geography	Combine LGA with ward, and region with district codes, to distinguish repeated names and represent codes as categories.
Validate age information	Treat nonpositive construction years and construction after recording as missing; derive pump age and recording year.
Represent seasonality	Encode recording month with sine and cosine, preserving adjacency between December and January.
Validate coordinates	Flag zero longitude, latitude within 10^{-6} degrees of zero, or incomplete coordinate pairs; mark both coordinates missing.
Represent local context	Add grid categories at 0.05-degree and 0.10-degree angular resolutions alongside continuous coordinates.
Handle zeros and skew	Retain zero indicators for population, elevation, and static head; log-transform population and static head after clipping negatives to zero.
Normalize category text	Standardize case and surrounding whitespace; convert empty strings to missing values.

2.1 Missing values and categorical encoding

Numeric variables use median imputation with indicators for missingness observed during fitting. Categorical missing values receive an explicit category followed by one-hot encoding. Categories appearing fewer than 20 times in a training fold are grouped as infrequent. Geographic categories have no additional cap; funder and installer are capped at 300 encoded categories each, and other categorical fields at 50.

Unseen categories map to the infrequent group when available; otherwise their encoded indicators are zero. KNN and logistic regression additionally standardize numeric predictors; tree models use unscaled inputs. Detailed and broad source and extraction categories are retained together where they provide complementary information.

2.2 Rationale and validation discipline

Coordinate grids represent angular rather than fixed physical distances. Zero indicators retain potentially informative zeros instead of treating all zeros as missing. The variable `amount_tsh` denotes total static head in the dataset description [1].

Row-level transformations are deterministic. Learned imputation, category grouping, encoding, and scaling are fitted within each training fold, so validation records do not determine those preprocessing statistics. Dropping detailed fields and grouping rare categories can nevertheless remove useful local signal.

3 Model selection and cross-validation

3.1 Validation design and metrics

A stratified random split with seed 42 reserves 5,940 records (10%) for holdout evaluation. The remaining 53,460 records use five shuffled stratified folds, each with 42,768 training and 10,692 validation records. All model families use the same folds, which approximately preserve class proportions [2].

Macro-F1 gives each class equal contribution, with class precision P_k and recall R_k:

$$F_{1,k} = \frac{2P_kR_k}{P_k + R_k}, \qquad F_{1,\text{macro}} = \frac{1}{3}\sum_{k=1}^{3} F_{1,k}. \tag{1}$$

Accuracy and repair recall are also reported. Grid search selects each family's configuration by mean macro-F1 and refits it on all development records; accuracy breaks ties between families. The grid contains 23 configurations, requiring 115 validation fits and eight refits. Earlier development experiments informed the grid, so these choices remain part of tuning.

3.2 Search scope and training weights

KNN tests 5/11 neighbors; logistic regression tests $C = 0.3/1.0$; the decision tree tests depth 20/unlimited. Random forest and Extra Trees test feature fractions 0.3/0.5 and leaf sizes 1/2, using 150 and 300 trees respectively. Bagging uses 300 trees and tests sample fractions 0.7/1.0 and feature fractions 0.5/0.75. XGBoost tests 750/1,000 rounds and learning rates 0.05/0.1 at depth 10.

Logistic regression, the decision tree, and bagging base trees use balanced class weights $w_k = n/(3n_k)$, recalculated within each training fit. Random forest, Extra Trees, and XGBoost use unweighted training; validation metrics are unweighted throughout. KNN uses distance weighting, which controls neighbor influence rather than class importance.

Table 3: Five-fold performance under argmax. SD is the standard deviation of macro-F1 across folds.

Model	Accuracy	Macro-F1	SD	Repair recall
Bagging trees	80.30%	0.7066	0.0027	46.33%
Extra Trees	80.61%	0.6973	0.0027	36.16%
Random forest	81.15%	0.6968	0.0033	33.64%
XGBoost	80.76%	0.6927	0.0044	33.98%
KNN	77.54%	0.6703	0.0020	37.63%
Decision tree	75.56%	0.6484	0.0051	38.76%
Logistic regression	69.61%	0.6229	0.0038	71.45%
Majority baseline	54.31%	0.2346	0.000013	0.00%

3.3 Interpretation

Bagging leads macro-F1 under argmax (Table 3), while random forest has the highest accuracy. Logistic regression attains the highest repair recall but lower overall macro-F1. Bagging's selected sample and feature fractions are 1.0 and 0.5, with minimum base-tree leaf size 2. Its 0.0093 macro-F1 advantage over Extra Trees is descriptive: fold variation is not a confidence interval, and these rankings do not establish statistical significance.

4 Threshold selection and holdout results

4.1 Selected decision rule

The top four families enter threshold selection. Each is fitted on the first fold's training records and scored on its 10,692 validation records using argmax and 91 repair thresholds from 0.05 to 0.95. The threshold rule predicts repair when $p_1 \geq t$; otherwise it selects the more probable of functional and non-functional. Rules are ranked by macro-F1, accuracy, repair recall, then preference for argmax [3].

Table 4: Best repair-threshold rules on the first validation fold; these are tuning scores.

Model	Threshold	Macro-F1	Accuracy	Repair recall
XGBoost	0.27	0.7067	80.13%	48.52%
Bagging trees	0.52	0.7048	80.88%	39.38%
Random forest	0.34	0.7040	80.62%	41.44%
Extra Trees	0.40	0.7001	80.36%	40.54%

XGBoost with threshold 0.27 is selected (Table 4), increasing first-fold macro-F1 from 0.6961 under argmax to 0.7067. It uses 1,000 rounds, learning rate 0.1, depth 10, minimum child weight 2, gamma 0.1, row/column sampling 0.9, and L1/L2 regularization 0.1/0.5. Training is unweighted, with a multiclass probability objective and histogram-based trees [4]. There is no early stopping. The rule is applied to the model refitted on all development records.

4.2 Holdout performance

The frozen model and rule correctly classify 4,822 of 5,940 holdout records: accuracy is 81.18% and macro-F1 0.7144. The majority baseline achieves 54.31% and 0.2346 respectively, giving improvements of 26.87 percentage points in accuracy and 0.4798 in macro-F1.

Table 5: Class-specific holdout performance.

Class	Precision	Recall	F1	Support
Functional	81.83%	88.38%	0.8498	3,226
Functional needs repair	46.71%	47.69%	0.4719	432
Non-functional	87.59%	77.34%	0.8215	2,282

Table 6: Holdout confusion matrix: actual condition in rows, predictions in columns.

	Functional	Needs repair	Non-functional
Functional	2,851	176	199
Needs repair	175	206	51
Non-functional	458	59	1,765

Repair detection is weakest (Table 5): only 206 of 432 repair cases receive the correct label. Of the remainder, 175 are predicted functional and 51 non-functional (Table 6). The latter still trigger inspection, illustrating why attention-case coverage differs from repair-class recall.

5 Inspection prioritization

5.1 Policy and retrospective outcomes

The illustrative policy selects waterpoints predicted repair or non-functional for inspection and skips those predicted functional. Actual repair and non-functional sites are attention cases. This evaluates workload and coverage using recorded condition labels.

Table 7: Retrospective outcomes of the inspection policy on 5,940 holdout sites.

Measure	Count	Share
Sites selected for inspection	2,456 of 5,940	41.35%
Attention cases selected	2,081 of 2,714	76.68%
Attention cases missed	633 of 2,714	23.32%
Functional sites selected	375 of 3,226	11.62%
Attention cases among selected sites	2,081 of 2,456	84.73%

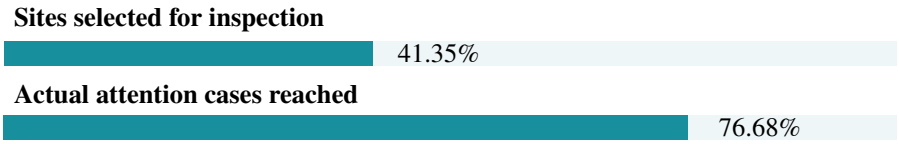


Figure 1: Workload and attention-case coverage, using all sites and actual attention cases as their respective denominators.

5.2 Workload–coverage trade-off

The policy selects 41.35% of sites and reaches 76.68% of attention cases (Table 7; Figure 1). Relative to inspecting every holdout site, it schedules 3,484 fewer visits, a reduction of 58.65%. Inspection yield is 84.73%, compared with an attention-case prevalence of 45.69%; the selected set is therefore substantially enriched for sites needing attention.

Reduced workload comes with 633 missed attention cases: 175 repair cases and 458 non-functional sites are predicted functional. A further 51 repair cases are predicted non-functional and 59 non-functional sites are predicted repair; these incorrect condition labels still trigger visits. Among actually functional sites, 176 are predicted repair and 199 non-functional, producing 375 unnecessary selections.

5.3 Operational interpretation

Three-class macro-F1 and inspection coverage describe different objectives. An acceptable policy depends on inspection capacity and the consequences of missed sites. The present rule cannot replace universal inspection where very high coverage is required; alternative rules should be compared using locally justified miss costs and operational capacity.

Predictors must also be available before deciding whether to inspect a site. Water quantity and operating characteristics may require field observation, reducing the potential benefit of using them to schedule visits. The workload reduction is a retrospective estimate of visit count, rather than measured financial savings or maintenance improvement.

6 Discussion and conclusion

6.1 Validation limitations

Hyperparameter and threshold selection share validation labels, and earlier experiments informed the search grid. Threshold scores are therefore tuning results. XGBoost's advantage over bagging after threshold adjustment is approximately 0.0019 macro-F1 on one fold, which does not establish consistent superiority. Holdout results were also reviewed during development, making fresh confirmation data important.

Random splits may place nearby waterpoints in both training and validation sets. Transfer across regions or collection periods remains uncertain. Geographic grouping and temporal validation would provide stronger evidence for new settings. Removing detailed fields and grouping rare categories may also discard useful information.

Predicted probabilities have not been separately assessed for calibration. The repair threshold should be interpreted as an empirically tuned decision rule, rather than a calibrated estimate of operational risk.

6.2 Further evaluation

A stronger selection procedure would separate threshold validation from hyperparameter search or use nested out-of-fold predictions, followed by evaluation of the frozen process on fresh data. Further analysis should examine repair cases predicted functional and assess performance across regions, collection periods, and site characteristics.

A prospective field pilot could compare prioritized visits with the existing inspection process, verify conditions at selected sites and a sample of unflagged sites, and track subsequent repair outcomes. Local inspection costs, travel constraints, predictor availability, and acceptable missed-case rates should guide policy design.

6.3 Conclusion

Eight classifiers are compared for three-class waterpoint condition prediction. Bagging leads five-fold macro-F1 under argmax, while repair-threshold selection chooses XGBoost with threshold 0.27. The selected model achieves 81.18% holdout accuracy and macro-F1 of 0.7144, substantially exceeding the majority baseline.

The inspection policy reaches 76.68% of attention cases while selecting 41.35% of sites, with 84.73% yield. Repair recall of 47.69% and 633 missed attention cases remain the principal limitations. The findings support further investigation of prediction-assisted inspection, while independent validation and field evaluation are needed to establish operational benefit. Recorded condition labels do not establish future failure risk, water safety, or maintenance improvement.

References

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